

Computer Graphics and Design

Module 4: Geometric Dimensioning and Tolerances

MCQ's

Introduction to Geometric Dimensioning and Tolerances

1. What is the main purpose of Geometric Dimensioning and Tolerancing (GD&T)?

- A) To increase manufacturing costs
- B) To define size, form, orientation, and location of features
- C) To color engineering drawings
- D) To replace engineering materials

Correct Answer: B) To define size, form, orientation, and location of features

Explanation:

GD&T is a symbolic language used in engineering drawings to clearly define the allowable variation in geometry and dimensions of parts.

2. Which standard is commonly used for GD&T?

- A) ISO 9001
- B) ASME Y14.5
- C) ASTM A36
- D) IEEE 802

Correct Answer: B) ASME Y14.5

Explanation:

ASME Y14.5 is the widely accepted standard that provides rules and symbols for GD&T.

3. What does a datum represent in GD&T?

- A) A machining process
- B) A reference point, axis, or surface

C) A type of material

D) A measurement instrument

Correct Answer: B) A reference point, axis, or surface

Explanation:

A datum is an exact reference used to establish measurement and inspection locations.

4. Which symbol is used for straightness in GD&T?

A) Circle

B) Square

C) Straight line

D) Triangle

Correct Answer: C) Straight line

Explanation:

The straightness symbol controls how much a feature may deviate from a perfect straight line.

5. GD&T mainly helps in:

A) Improving communication in engineering drawings

B) Increasing product weight

C) Reducing machine speed

D) Changing material properties

Correct Answer: A) Improving communication in engineering drawings

Explanation:

GD&T provides a universal engineering language that improves clarity between design, manufacturing, and inspection teams.

6. Which GD&T characteristic controls the roundness of a feature?

A) Flatness

- B) Circularity
- C) Parallelism
- D) Symmetry

Correct Answer: B) Circularity

Explanation:

Circularity ensures that every circular element of a feature remains within specified tolerance limits.

7. In GD&T, the feature control frame is used to:

- A) Show material type
- B) Specify geometric tolerances
- C) Define heat treatment
- D) Indicate surface color

Correct Answer: B) Specify geometric tolerances

Explanation:

The feature control frame contains symbols, tolerance values, and datum references related to geometric controls.

8. Which tolerance controls the flatness of the surface?

- A) Flatness tolerance
- B) Profile tolerance
- C) Runout tolerance
- D) Angularity tolerance

Correct Answer: A) Flatness tolerance

Explanation:

Flatness tolerance limits the variation of a surface from an ideal flat plane.

9. Which of the following is an orientation control in GD&T?

- A) Straightness
- B) Circularity
- C) Perpendicularity
- D) Position

Correct Answer: C) Perpendicularity

Explanation:

Perpendicularity controls the orientation of a feature at 90° relative to a datum.

10. What is the advantage of using GD&T over traditional tolerancing?

- A) It increases ambiguity
- B) It reduces manufacturing accuracy
- C) It provides better functional control of parts
- D) It eliminates the need for dimensions

Correct Answer: C) It provides better functional control of parts

Explanation:

GD&T ensures that parts fit and function properly while allowing reasonable manufacturing variations.

11. Which GD&T control is most suitable for controlling the exact location of a hole?

- A) Flatness
- B) Position
- C) Circularity
- D) Straightness

Correct Answer: B) Position

Explanation:

Position tolerance precisely controls the allowable variation in the location of features such as holes.

12. In GD&T, Maximum Material Condition (MMC) refers to:

- A) The condition with minimum material
- B) The average material condition
- C) The condition where a feature contains the maximum amount of material
- D) The condition of zero tolerance

Correct Answer: C) The condition where a feature contains the maximum amount of material

Explanation:

MMC occurs when a feature is at its maximum material size, such as the largest shaft or smallest hole.

13. Which geometric characteristic belongs to the form tolerance category?

- A) Parallelism
- B) Position
- C) Cylindricity
- D) Angularity

Correct Answer: C) Cylindricity

Explanation:

Cylindricity controls the overall form of a cylindrical feature and belongs to form tolerances.

14. A datum reference frame in GD&T is established using:

- A) Colors and notes
- B) One or more datums
- C) Surface roughness values
- D) Machining parameters

Correct Answer: B) One or more datums

Explanation:

A datum reference frame is created from specified datums to provide a consistent coordinate system for measurement.

15. Which statement about GD&T is correct?

- A) GD&T only controls dimensions
- B) GD&T replaces all measurements
- C) GD&T defines allowable geometric variation in manufactured parts
- D) GD&T is only used for casting operations

Correct Answer: C) GD&T defines allowable geometric variation in manufactured parts

Explanation:

GD&T specifies acceptable geometric variations to ensure proper assembly, interchangeability, and functionality of parts.

Importance of dimensioning and its rules

16. What is the main purpose of dimensioning in engineering drawings?

- A) To decorate the drawing
- B) To specify the size and location of features
- C) To increase drawing thickness
- D) To reduce material strength

Correct Answer: B) To specify the size and location of features

Explanation:

Dimensioning provides clear information about the size, shape, and position of features required for manufacturing and inspection.

17. Which of the following is a basic rule of dimensioning?

- A) Dimensions should be repeated many times
- B) Dimensions should be placed clearly and neatly
- C) Dimensions should overlap each other
- D) Dimensions should be written randomly

Correct Answer: B) Dimensions should be placed clearly and neatly

Explanation:

Clear and neat placement of dimensions improves readability and avoids confusion during manufacturing.

18. Why is proper dimensioning important in manufacturing?

- A) It increases machine noise
- B) It ensures accurate production of parts
- C) It changes material properties
- D) It reduces drawing size

Correct Answer: B) It ensures accurate production of parts

Explanation:

Correct dimensioning helps manufacturers produce components with the required accuracy and functionality.

19. In engineering drawings, dimensions are usually expressed in:

- A) Colors
- B) Symbols only
- C) Numerical values
- D) Pictures

Correct Answer: C) Numerical values

Explanation:

Dimensions are represented using numerical values along with units and symbols for precise communication.

20. Which line is mainly used for showing dimensions?

- A) Border line
- B) Dimension line
- C) Center line

D) Object line

Correct Answer: B) Dimension line

Explanation:

Dimension lines indicate the direction and extent of a dimension in a drawing.

21. Which rule should be followed while dimensioning a drawing?

A) Avoid unnecessary dimensions

B) Repeat dimensions frequently

C) Place dimensions inside object lines only

D) Use different units in one drawing

Correct Answer: A) Avoid unnecessary dimensions

Explanation:

Unnecessary dimensions can create confusion and increase the chances of errors.

22. What is the importance of standardized dimensioning rules?

A) They reduce material hardness

B) They improve communication between engineers and manufacturers

C) They increase drawing costs

D) They eliminate measurements

Correct Answer: B) They improve communication between engineers and manufacturers

Explanation:

Standardized rules ensure everyone interprets the drawing in the same way.

23. Which of the following should generally not be done in dimensioning?

A) Provide sufficient dimensions

B) Keep dimensions readable

C) Cross dimension lines unnecessarily

D) Use standard symbols

Correct Answer: C) Cross dimension lines unnecessarily

Explanation:

Crossing dimension lines makes drawings difficult to read and should be minimized.

24. The dimensions of a drawing should primarily be based on:

A) Appearance of the drawing

B) Manufacturing and functional requirements

C) Color selection

D) Weight of the paper

Correct Answer: B) Manufacturing and functional requirements

Explanation:

Dimensions must support proper manufacturing, assembly, and functioning of the component.

25. Which method places dimensions in line with the dimension line so they can be read from the bottom or right side?

A) Chain dimensioning

B) Aligned method

C) Baseline dimensioning

D) Coordinate dimensioning

Correct Answer: B) Aligned method

Explanation:

In the aligned method, dimension values are placed parallel to dimension lines and read from the bottom or right side.

26. Why should dimensions not be duplicated in engineering drawings?

A) It increases paper usage

B) It may create conflicts and confusion

- C) It changes material properties
- D) It improves drawing quality

Correct Answer: B) It may create conflicts and confusion

Explanation:

Duplicate dimensions can lead to inconsistencies and manufacturing errors if values do not match.

27. Which dimensioning practice best supports interchangeability of parts?

- A) Random dimension placement
- B) Functional dimensioning
- C) Decorative dimensioning
- D) Approximate dimensioning

Correct Answer: B) Functional dimensioning

Explanation:

Functional dimensioning ensures that manufactured parts fit and work correctly with mating components.

28. In proper dimensioning practice, dimensions should be given:

- A) According to the convenience of the drafter only
- B) According to manufacturing and inspection needs
- C) Without considering tolerance
- D) Only on isometric views

Correct Answer: B) According to manufacturing and inspection needs

Explanation:

Dimensions must help both manufacturing and quality inspection processes.

29. What is the main disadvantage of poor dimensioning?

- A) Improved readability

- B) Increased production accuracy
- C) Manufacturing and assembly errors
- D) Reduced drawing time

Correct Answer: C) Manufacturing and assembly errors

Explanation:

Incorrect or unclear dimensions can result in defective parts and assembly problems.

30. Which statement is correct regarding dimensioning rules?

- A) Every feature should be dimensioned more than once
- B) Hidden lines should preferably not be dimensioned
- C) Dimensions can be placed randomly
- D) Units should change frequently within the drawing

Correct Answer: B) Hidden lines should preferably not be dimensioned

Explanation:

Dimensioning hidden lines reduces clarity. Dimensions should preferably be placed on visible features whenever possible.

Tolerances, Limits, fits and its types

31. What is meant by tolerance in engineering?

- A) Exact size of a part
- B) Difference between upper and lower limits
- C) Maximum size only
- D) Minimum size only

Correct Answer: B) Difference between upper and lower limits

Explanation:

Tolerance is the permissible variation in a dimension and is calculated as the difference between maximum and minimum limits.

32. The maximum allowable size of a part is called:

- A) Basic size
- B) Lower limit
- C) Upper limit
- D) Tolerance

Correct Answer: C) Upper limit

Explanation:

The upper limit is the highest permissible dimension of a component.

33. Which fit provides always a gap between shaft and hole?

- A) Interference fit
- B) Transition fit
- C) Clearance fit
- D) Tight fit

Correct Answer: C) Clearance fit

Explanation:

In a clearance fit, the hole size is always larger than the shaft size, ensuring free movement.

34. What is the lower limit of a dimension?

- A) Largest permissible size
- B) Average size
- C) Smallest permissible size
- D) Basic size

Correct Answer: C) Smallest permissible size

Explanation:

The lower limit is the minimum acceptable dimension for a component.

35. Which type of fit may result in either clearance or interference?

- A) Clearance fit
- B) Transition fit
- C) Interference fit
- D) Running fit

Correct Answer: B) Transition fit

Explanation:

Transition fits may provide either a small clearance or slight interference depending on actual dimensions.

36. A shaft has limits of 24.98 mm and 25.02 mm. What is tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: C) 0.04 mm

Explanation:

Tolerance = Upper limit – Lower limit
= 25.02 – 24.98 = 0.04 mm

37. Which fit is generally used for permanent assembly?

- A) Clearance fit
- B) Transition fit
- C) Interference fit
- D) Running fit

Correct Answer: C) Interference fit

Explanation:

Interference fits are used where components are intended to remain firmly assembled.

38. In a hole-basis system, which feature remains constant?

- A) Shaft size
- B) Hole size
- C) Tolerance value
- D) Basic size of shaft only

Correct Answer: B) Hole size

Explanation:

In the hole-basis system, the hole size is kept constant while shaft size varies to obtain different fits.

39. Which of the following best defines limits?

- A) Exact dimensions of part
- B) Maximum and minimum permissible sizes
- C) Difference between sizes
- D) Surface finish values

Correct Answer: B) Maximum and minimum permissible sizes

Explanation:

Limits specify the allowable extreme sizes for a dimension.

40. A hole size varies from 40.00 mm to 40.05 mm. What is the tolerance of the hole?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: D) 0.05 mm

Explanation:

$$\text{Tolerance} = 40.05 - 40.00 = 0.05 \text{ mm}$$

41. Which fit is suitable for parts requiring accurate location but easy assembly?

- A) Interference fit
- B) Clearance fit
- C) Transition fit
- D) Loose fit

Correct Answer: C) Transition fit

Explanation:

Transition fits provide accurate alignment with limited clearance or interference.

42. A shaft diameter is 20.02 mm and the hole diameter is 20.00 mm. Which type of fit is obtained?

- A) Clearance fit
- B) Transition fit
- C) Interference fit
- D) Running fit

Correct Answer: C) Interference fit

Explanation:

Since the shaft is larger than the hole, interference occurs during assembly.

43. What is the allowance in a clearance fit?

- A) Minimum interference between hole and shaft
- B) Maximum interference between hole and shaft
- C) Minimum clearance between hole and shaft
- D) Difference between basic sizes only

Correct Answer: C) Minimum clearance between hole and shaft

Explanation:

Allowance is the intentional difference between mating parts and equals minimum clearance in clearance fits.

44. A hole has limits of 50.00 mm and 50.04 mm, while a shaft has limits of 49.96 mm and 49.98 mm. What is the maximum clearance?

A) 0.02 mm

B) 0.04 mm

C) 0.06 mm

D) 0.08 mm

Correct Answer: D) 0.08 mm

Explanation:

Maximum clearance = Maximum hole size – Minimum shaft size

= 50.04 – 49.96 = 0.08 mm

45. Which statement about tolerances is correct?

A) Zero tolerance is always practical in manufacturing

B) Tolerances allow permissible variation in dimensions

C) Tolerances eliminate inspection

D) Tolerances increase manufacturing errors

Correct Answer: B) Tolerances allow permissible variation in dimensions

Explanation:

Tolerances are necessary because exact dimensions cannot be achieved economically in manufacturing.

46. A shaft has dimensions 30 ± 0.02 mm. What are the upper and lower limits?

A) 30.01 mm and 29.99 mm

- B) 30.02 mm and 29.98 mm
- C) 30.04 mm and 29.96 mm
- D) 30.03 mm and 29.97 mm

Correct Answer: B) 30.02 mm and 29.98 mm

Explanation:

Upper limit = $30 + 0.02 = 30.02$ mm

Lower limit = $30 - 0.02 = 29.98$ mm

Symbolic representations, and their meaning for geometrical objects

47. What is the purpose of symbolic representation in engineering drawing?

- A) To decorate the drawing
- B) To provide standard information quickly and clearly
- C) To increase drawing size
- D) To replace dimensions completely

Correct Answer: B) To provide standard information quickly and clearly

Explanation:

Symbols help engineers communicate geometrical information in a simple and universally understood manner.

48. Which symbol is commonly used to represent diameter?

- A) $\square$
- B) R
- C) $\varnothing$
- D) $\perp$

Correct Answer: C) $\varnothing$

Explanation:

The symbol $\varnothing$ indicates the diameter of a circular feature.

49. What does the symbol “R” represent in engineering drawings?

- A) Roughness
- B) Radius
- C) Rotation
- D) Reference

Correct Answer: B) Radius

Explanation:

The symbol “R” is used before a dimension to indicate the radius of an arc or circle.

50. Which geometrical object is represented by the symbol “⊥”?

- A) Parallelism
- B) Circularity
- C) Perpendicularity
- D) Straightness

Correct Answer: C) Perpendicularity

Explanation:

The symbol “⊥” indicates that two surfaces or axes must be at 90° to each other.

51. In engineering graphics, the center line is generally represented by:

- A) Thick continuous line
- B) Dashed thick line
- C) Long dash-short dash thin line
- D) Zigzag line

Correct Answer: C) Long dash-short dash thin line

Explanation:

Center lines are used to represent axes and centers of circular objects.

52. Which symbol represents parallelism in GD&T?

- A) $\parallel$
- B) $\perp$
- C) $\bigcirc$
- D) ϕ

Correct Answer: A) $\parallel$

Explanation:

The parallelism symbol indicates that a feature must remain parallel to a datum reference.

53. The symbol “ ϕ ” in GD&T represents:

- A) Flatness
- B) Position
- C) Circularity
- D) Symmetry

Correct Answer: B) Position

Explanation:

The position symbol controls the exact location of a feature relative to datums.

54. Which symbol is used to indicate surface roughness?

- A) V-shaped symbol
- B) Circle
- C) Triangle
- D) Rectangle

Correct Answer: A) V-shaped symbol

Explanation:

A surface finish symbol is commonly represented by a check-mark or V-shaped symbol.

55. What does the symbol “○” represent in GD&T?

- A) Circularity
- B) Cylindricity
- C) Diameter
- D) Radius

Correct Answer: A) Circularity

Explanation:

Circularity controls how closely a feature approaches a perfect circle.

56. Which line type is used to represent hidden geometrical objects?

- A) Continuous thick line
- B) Continuous thin line
- C) Dashed thin line
- D) Chain thick line

Correct Answer: C) Dashed thin line

Explanation:

Hidden lines show edges or surfaces that are not directly visible in a view.

57. Which GD&T symbol controls the orientation of a feature at an angle other than 90°?

- A) Parallelism
- B) Straightness
- C) Angularity
- D) Circularity

Correct Answer: C) Angularity

Explanation:

Angularity controls the inclination of a feature relative to a datum at a specified angle.

58. The symbol for flatness in GD&T is used to control:

- A) Roundness of a surface
- B) Straightness of an axis
- C) Uniform flatness of a surface
- D) Position of holes

Correct Answer: C) Uniform flatness of a surface

Explanation:

Flatness ensures that a surface lies within two parallel planes.

59. Which symbol is associated with concentricity in GD&T?

- A) $\odot$
- B) $\parallel$
- C) $\perp$
- D) $\frown$

Correct Answer: A) $\odot$

Explanation:

The concentricity symbol controls the alignment of the center points of circular features.

60. What is the meaning of the datum symbol in engineering drawings?

- A) Surface finish requirement
- B) Reference feature for measurement
- C) Hidden feature location
- D) Material specifications

Correct Answer: B) Reference feature for measurement

Explanation:

A datum provides a reference point, line, or surface for dimensioning and tolerancing.

61. Which statement about symbolic representation is correct?

- A) Symbols increase ambiguity in drawings
- B) Symbols are standardized for universal understanding
- C) Symbols are only used in machine design
- D) Symbols replace all written notes

Correct Answer: B) Symbols are standardized for universal understanding

Explanation:

Engineering symbols follow international standards to ensure clear communication among engineers, manufacturers, and inspectors.

Practical application and interpretation of GD &T using computer graphics

62. What is the primary use of GD&T in computer graphics-based design software?

- A) To change drawing colors
- B) To define and control geometric features accurately
- C) To increase file size
- D) To replace CAD software

Correct Answer: B) To define and control geometric features accurately

Explanation:

GD&T in CAD software helps designers specify the allowable variation in geometry and dimensions of parts.

63. Which type of software is commonly used for applying GD&T in computer graphics?

- A) Word processing software
- B) Spreadsheet software
- C) CAD software
- D) Video editing software

Correct Answer: C) CAD software

Explanation:

Computer-Aided Design (CAD) software is widely used for creating and interpreting GD&T-based engineering drawings.

64. In CAD drawings, GD&T symbols are generally placed inside:

- A) Object lines
- B) Feature control frames
- C) Border lines
- D) Center lines

Correct Answer: B) Feature control frames

Explanation:

Feature control frames contain geometric tolerance symbols, values, and datum references.

65. Which GD&T concept helps ensure proper assembly of parts in computer-modeled designs?

- A) Color coding
- B) Animation
- C) Tolerancing
- D) Rendering

Correct Answer: C) Tolerancing

Explanation:

Tolerancing ensures parts fit and function properly during assembly.

66. What does CAD stand for?

- A) Computer Aided Drafting/Design
- B) Controlled Assembly Design
- C) Computerized Axis Drawing
- D) Calculated Angular Design

Correct Answer: A) Computer Aided Drafting/Design

Explanation:

CAD refers to software used for creating, modifying, and analyzing engineering drawings and models.

67. Which GD&T symbol is commonly used in CAD software to control hole location?

- A) Flatness
- B) Position
- C) Circularity
- D) Straightness

Correct Answer: B) Position

Explanation:

The position tolerance controls the exact location of holes and other features.

68. In computer graphics-based GD&T interpretation, datums are mainly used for:

- A) Coloring surfaces
- B) Providing reference points for measurements
- C) Increasing model speed
- D) Reducing file size

Correct Answer: B) Providing reference points for measurements

Explanation:

Datums establish reference planes, axes, or points for accurate dimensioning and inspection.

69. Which benefit is achieved by using GD&T in 3D CAD models?

- A) Increased ambiguity
- B) Better communication between design and manufacturing teams
- C) Elimination of measurements

D) Reduced visualization quality

Correct Answer: B) Better communication between design and manufacturing teams

Explanation:

GD&T provides a standardized method for communicating design intent clearly.

70. Which CAD feature allows engineers to inspect dimensions and tolerances digitally?

A) Rendering tool

B) Simulation and measurement tools

C) Audio settings

D) Printing options

Correct Answer: B) Simulation and measurement tools

Explanation:

CAD software includes measurement and simulation tools for checking dimensions and tolerances virtually.

71. What is the role of feature control frames in computer graphics interpretation of GD&T?

A) To display colors

B) To store images

C) To specify geometric tolerance information

D) To increase drawing scale

Correct Answer: C) To specify geometric tolerance information

Explanation:

Feature control frames communicate tolerance type, tolerance value, and datum references.

72. Which practical advantage does GD&T provide in computer-aided manufacturing (CAM)?

A) Reduces machine lubrication

B) Improves manufacturing accuracy and consistency

- C) Eliminates CAD models
- D) Increases material waste

Correct Answer: B) Improves manufacturing accuracy and consistency

Explanation:

GD&T ensures precise interpretation of design requirements during manufacturing.

73. In a CAD environment, virtual inspection of GD&T helps to:

- A) Increase drawing thickness
- B) Detect tolerance-related assembly problems before production
- C) Reduce computer memory
- D) Eliminate datums

Correct Answer: B) Detect tolerance-related assembly problems before production

Explanation:

Virtual inspection helps identify fit and alignment issues before actual manufacturing begins.

74. Which statement best describes the interpretation of GD&T in computer graphics?

- A) It depends only on visual appearance
- B) It uses standard symbols and rules for geometric control
- C) It ignores manufacturing requirements
- D) It replaces dimensions completely

Correct Answer: B) It uses standard symbols and rules for geometric control

Explanation:

GD&T interpretation follows standardized symbols and principles within CAD systems.

75. Why are 3D CAD models preferred for practical GD&T applications?

- A) They reduce geometric accuracy
- B) They allow better visualization and analysis of tolerances

- C) They avoid engineering standards
- D) They eliminate the need for measurements

Correct Answer: B) They allow better visualization and analysis of tolerances

Explanation:

3D models help engineers understand feature relationships and tolerance effects more effectively.

76. Which process commonly uses GD&T information extracted from computer graphics models?

- A) Painting process
- B) Quality inspection and CNC machining
- C) Packaging design only
- D) Manual sketching only

Correct Answer: B) Quality inspection and CNC machining

Explanation:

GD&T data from CAD models is widely used in CNC machining and quality inspection for accurate manufacturing and verification.

Numerical on GD&T, MMC, LMC etc.

77. A shaft has a size of 20 ± 0.02 mm. What is the Maximum Material Condition (MMC) of the shaft?

- A) 19.98 mm
- B) 20.00 mm
- C) 20.02 mm
- D) 20.04 mm

Correct Answer: C) 20.02 mm

Explanation:

For a shaft, MMC is the largest size because it contains the maximum amount of material.

78. A hole is specified as 30 ± 0.05 mm. What is the Least Material Condition (LMC) of the hole?

- A) 29.95 mm
- B) 30.00 mm
- C) 30.05 mm
- D) 30.10 mm

Correct Answer: C) 30.05 mm

Explanation:

For a hole, LMC is the largest hole size because it contains the least material.

79. A hole has limits of 15.00 mm and 15.04 mm. What is tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: C) 0.04 mm

Explanation:

Tolerance = Upper limit – Lower limit

= $15.04 - 15.00 = 0.04$ mm

80. For a shaft, which condition represents MMC?

- A) Smallest diameter
- B) Average diameter
- C) Largest diameter
- D) Zero diameter

Correct Answer: C) Largest diameter

Explanation:

A larger shaft contains more material; therefore, it represents MMC.

81. A shaft dimension is 25 ± 0.01 mm. What is the LMC size?

- A) 24.99 mm
- B) 25.00 mm
- C) 25.01 mm
- D) 25.02 mm

Correct Answer: A) 24.99 mm

Explanation:

For shafts, LMC is the smallest size because it contains the least material.

82. A hole has dimensions 40.00 mm to 40.08 mm. What is the MMC of the hole?

- A) 39.92 mm
- B) 40.00 mm
- C) 40.04 mm
- D) 40.08 mm

Correct Answer: B) 40.00 mm

Explanation:

For holes, MMC is the smallest hole size because it contains the maximum material.

83. A shaft has limits of 10.02 mm and 9.98 mm. What is total tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: C) 0.04 mm

Explanation:

Tolerance = $10.02 - 9.98 = 0.04$ mm

84. A positional tolerance of $\varnothing 0.10$ mm at MMC is applied to a hole. If the actual hole size departs from MMC by 0.03 mm, what is the bonus tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.05 mm
- D) 0.10 mm

Correct Answer: B) 0.03 mm

Explanation:

Bonus tolerance equals the amount by which the feature departs from MMC.

85. Which statement is correct regarding MMC?

- A) MMC always gives minimum material
- B) MMC is independent of feature size
- C) MMC represents the condition with maximum material
- D) MMC applies only to holes

Correct Answer: C) MMC represents the condition with maximum material

Explanation:

MMC defines the feature size containing the greatest amount of material.

86. A hole has a diameter tolerance from 12.00 mm to 12.06 mm. What is LMC?

- A) 12.00 mm
- B) 12.03 mm
- C) 12.06 mm
- D) 11.94 mm

Correct Answer: C) 12.06 mm

Explanation:

For holes, the largest size corresponds to LMC.

87. A shaft is specified as 50 ± 0.05 mm. What is the MMC and LMC respectively?

- A) 49.95 mm and 50.05 mm
- B) 50.05 mm and 49.95 mm
- C) 50.00 mm and 49.95 mm
- D) 50.05 mm and 50.00 mm

Correct Answer: B) 50.05 mm and 49.95 mm

Explanation:

For shafts:

MMC = Largest size = 50.05 mm

LMC = Smallest size = 49.95 mm

88. A hole has MMC size 20.00 mm and positional tolerance $\varnothing 0.20$ mm at MMC. If the produced hole size is 20.08 mm, what is the total positional tolerance available?

- A) 0.20 mm
- B) 0.24 mm
- C) 0.28 mm
- D) 0.30 mm

Correct Answer: C) 0.28 mm

Explanation:

Bonus tolerance = $20.08 - 20.00 = 0.08$ mm

Total tolerance = $0.20 + 0.08 = 0.28$ mm

89. A shaft has dimensions ranging from 29.94 mm to 30.00 mm. What is the MMC?

- A) 29.94 mm

- B) 29.97 mm
- C) 30.00 mm
- D) 30.06 mm

Correct Answer: C) 30.00 mm

Explanation:

For shafts, MMC is the largest permissible size.

90. A hole tolerance is given as $18 +0.04/0$ mm. What is the tolerance value?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: C) 0.04 mm

Explanation:

Tolerance = Upper limit – Lower limit

$$= 18.04 - 18.00 = 0.04 \text{ mm}$$

91. A positional tolerance of $\varnothing 0.15$ mm at MMC is specified for a shaft. The shaft MMC is 25.05 mm, but actual shaft size is 25.00 mm. What is bonus tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.05 mm
- D) 0.10 mm

Correct Answer: C) 0.05 mm

Explanation:

Bonus tolerance = MMC size – Actual size

$$= 25.05 - 25.00 = 0.05 \text{ mm for a shaft feature.}$$

92. A hole size is given as 35 ± 0.03 mm. What is the MMC of the hole?

- A) 34.97 mm
- B) 35.00 mm
- C) 35.03 mm
- D) 35.06 mm

Correct Answer: A) 34.97 mm

Explanation:

For holes, MMC is the smallest size because it contains the maximum material.

93. A shaft has dimensions from 19.95 mm to 20.00 mm. What is tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: D) 0.05 mm

Explanation:

Tolerance = $20.00 - 19.95 = 0.05$ mm

94. Which condition gives the least amount of material in a shaft?

- A) MMC
- B) LMC
- C) Basic size
- D) Datum size

Correct Answer: B) LMC

Explanation:

For shafts, LMC is the minimum diameter condition.

95. A hole has a positional tolerance of $\varnothing 0.12$ mm at MMC. If the actual hole size is larger than MMC by 0.04 mm, what is the total positional tolerance?

- A) 0.12 mm
- B) 0.14 mm
- C) 0.16 mm
- D) 0.18 mm

Correct Answer: C) 0.16 mm

Explanation:

Total positional tolerance = Given tolerance + Bonus tolerance
 $= 0.12 + 0.04 = 0.16$ mm

96. A shaft size is specified as $45_{-0.02}$ mm. What is the LMC of the shaft?

- A) 44.96 mm
- B) 44.98 mm
- C) 45.00 mm
- D) 45.02 mm

Correct Answer: B) 44.98 mm

Explanation:

LMC for a shaft is the smallest size.

$45 - 0.02 = 44.98$ mm

97. A hole varies from 10.00 mm to 10.05 mm. Which value represents LMC?

- A) 10.00 mm
- B) 10.02 mm
- C) 10.03 mm
- D) 10.05 mm

Correct Answer: D) 10.05 mm

Explanation:

For holes, the largest size represents Least Material Condition.

98. A shaft has an MMC of 16.10 mm and actual size of 16.06 mm. What is bonus tolerance?

- A) 0.02 mm
- B) 0.03 mm
- C) 0.04 mm
- D) 0.05 mm

Correct Answer: C) 0.04 mm

Explanation:

Bonus tolerance = MMC – Actual size

$$= 16.10 - 16.06 = 0.04 \text{ mm}$$

99. A hole is specified as $25 +0.08/0$ mm with positional tolerance $\varnothing 0.10$ mm at MMC. If actual hole size is 25.05 mm, what is total positional tolerance?

- A) 0.10 mm
- B) 0.12 mm
- C) 0.15 mm
- D) 0.18 mm

Correct Answer: C) 0.15 mm

Explanation:

MMC of hole = 25.00 mm

$$\text{Bonus tolerance} = 25.05 - 25.00 = 0.05 \text{ mm}$$

$$\text{Total tolerance} = 0.10 + 0.05 = 0.15 \text{ mm}$$

100. A shaft is dimensioned as 60 ± 0.04 mm. What are the upper and lower limits?

- A) 60.02 mm and 59.98 mm

B) 60.04 mm and 59.96 mm

C) 60.06 mm and 59.94 mm

D) 60.08 mm and 59.92 mm

Correct Answer: B) 60.04 mm and 59.96 mm

Explanation:

Upper limit = $60 + 0.04 = 60.04$ mm

Lower limit = $60 - 0.04 = 59.96$ mm

101. A hole has limits of 50.00 mm and 50.10 mm, while a shaft has limits 49.95 mm and 50.02 mm. What is the maximum clearance?

A) 0.08 mm

B) 0.10 mm

C) 0.12 mm

D) 0.15 mm

Correct Answer: D) 0.15 mm

Explanation:

Maximum clearance = Largest hole – smallest shaft

= $50.10 - 49.95 = 0.15$ mm

102. Which GD&T symbol represents straightness?

A) ○

B) ⊥

C) —

D) ⊕

Correct Answer: C) —

Explanation:

The straight line symbol (—) represents straightness and controls the straight form of a feature.

103. What does the symbol “○” represent in GD&T?

- A) Flatness
- B) Circularity
- C) Parallelism
- D) Position

Correct Answer: B) Circularity

Explanation:

The circular symbol (○) represents circularity or roundness.

104. Which symbol is used for flatness?

- A) ▭
- B) ○
- C) ◎
- D) Ⓢ

Correct Answer: A) ▭

Explanation:

The parallelogram symbol (▭) represents flatness.

105. Which GD&T symbol represents position tolerance?

- A) ◎
- B) Ⓢ
- C) ○
- D) ∟

Correct Answer: B) Ⓢ

Explanation:

The position symbol (Ⓢ) controls the exact location of features.

106. What does the symbol “ $\angle$ ” indicate in GD&T?

- A) Circularity
- B) Angularity
- C) Cylindricity
- D) Runout

Correct Answer: B) Angularity

Explanation:

The angularity symbol controls orientation at a specified angle.

107. Which symbol represents cylindricity?

- A) ⌀
- B) $\bigcirc$
- C) $\perp$
- D) $\parallel$

Correct Answer: A) ⌀

Explanation:

The cylindricity symbol controls the complete cylindrical form of a feature.

108. The symbol “ $\frown$ ” is used for:

- A) Profile of a line
- B) Position
- C) Concentricity
- D) Symmetry

Correct Answer: A) Profile of a line

Explanation:

The curved line symbol ($\frown$) represents profile of a line.

109. Which symbol represents concentricity?

A) 

B) 

C) 

D) 

Correct Answer: A) 

Explanation:

The concentric circles symbol () controls concentricity between features.

110. The symbol “” represents:

A) Symmetry

B) Total runout

C) Straightness

D) Flatness

Correct Answer: B) Total runout

Explanation:

The symbol  is used for total runout tolerance.

111. Which pair of symbols belongs to orientation tolerances?

A)  and 

B)  and 

C)  and 

D)  and 

Correct Answer: B)  and 

Explanation:

Parallelism ($//$) and perpendicularity ($\perp$) are orientation controls.

112. Which symbol is associated with profile of a surface?

A) $\triangle$

B) ϕ

C) $\odot$

D) $\cup$

Correct Answer: A) $\triangle$

Explanation:

The symbol $\triangle$ represents profile of a surface and controls the contour of surfaces.

113. Which GD&T category does the straightness symbol belong to?

A) Orientation control

B) Form control

C) Runout control

D) Profile control

Correct Answer: B) Form control

Explanation:

Straightness controls the form of a feature and belongs to the form tolerance category.

114. The flatness symbol in GD&T is classified under:

A) Form tolerance

B) Orientation tolerance

C) Location tolerance

D) Runout tolerance

Correct Answer: A) Form tolerance

Explanation:

Flatness controls the shape of a surface without reference to datums.

115. Which symbol belongs to the orientation tolerance group?

- A) Circularity
- B) Cylindricity
- C) Parallelism
- D) Profile of surface

Correct Answer: C) Parallelism

Explanation:

Parallelism controls the orientation of a feature relative to a datum.

116. Circularity belongs to which GD&T category?

- A) Form tolerance
- B) Runout tolerance
- C) Location tolerance
- D) Orientation tolerance

Correct Answer: A) Form tolerance

Explanation:

Circularity controls the roundness of a feature and is a form tolerance.

117. Which GD&T category contains the symbols parallelism, perpendicularity, and angularity?

- A) Profile controls
- B) Orientation controls
- C) Form controls
- D) Runout controls

Correct Answer: B) Orientation controls

Explanation:

These symbols control the orientation of features relative to datums.

118. The position symbol “⌀” belongs to which tolerance category?

- A) Form
- B) Profile
- C) Location
- D) Runout

Correct Answer: C) Location

Explanation:

Position is a location tolerance used to control feature location.

119. Which of the following is profile tolerance?

- A) Straightness
- B) Profile of a surface
- C) Parallelism
- D) Circular runout

Correct Answer: B) Profile of a surface

Explanation:

Profile tolerances control the shape of lines and surfaces.

120. Concentricity and symmetry are classified under:

- A) Location tolerances
- B) Orientation tolerances
- C) Form tolerances
- D) Profile tolerances

Correct Answer: A) Location tolerances

Explanation:

Both concentricity and symmetry control the location relationship between features.

121. Which GD&T category includes circular runout and total runout?

- A) Form control
- B) Orientation control
- C) Runout control
- D) Profile control

Correct Answer: C) Runout control

Explanation:

Runout tolerances control the variation of rotating features.

122. How many major categories are commonly used to classify the 14 GD&T symbols?

- A) 3
- B) 4
- C) 5
- D) 6

Correct Answer: C) 5

Explanation:

The 14 GD&T symbols are grouped into five categories: Form, Orientation, Location, Profile, and Runout.

123. Which set correctly represents all form tolerances?

- A) Straightness, Flatness, Circularity, Cylindricity
- B) Parallelism, Perpendicularity, Angularity
- C) Position, Symmetry, Concentricity
- D) Circular Runout, Total Runout, Flatness

Correct Answer: A) Straightness, Flatness, Circularity, Cylindricity

Explanation:

These four symbols belong exclusively to form tolerances.

124. Which GD&T symbol controls the overall cylindrical shape of a feature?

- A) Circularity
- B) Cylindricity
- C) Straightness
- D) Flatness

Correct Answer: B) Cylindricity

Explanation:

Cylindricity controls the complete cylindrical form including roundness and straightness.

125. The profile of a line and profile of a surface symbols belong to which category?

- A) Profile tolerances
- B) Location tolerances
- C) Form tolerances
- D) Runout tolerances

Correct Answer: A) Profile tolerances

Explanation:

These tolerances control the contour of lines and surfaces.

126. Which statement about GD&T symbols is correct?

- A) All GD&T symbols require datums
- B) Form tolerances generally do not require datums
- C) Runout tolerances do not control rotation
- D) Orientation tolerances are independent of datums

Correct Answer: B) Form tolerances generally do not require datums

Explanation:

Form tolerances like straightness and flatness usually control individual features without datum references.

127. Which of the following symbols belongs to the runout category?

- A) Angularity
- B) Circular runout
- C) Position
- D) Circularity

Correct Answer: B) Circular runout

Explanation:

Circular runout controls variation during rotation and belongs to runout tolerances.

128. The angularity symbol controls:

- A) Form of a feature
- B) Orientation at a specified angle
- C) Hole location
- D) Surface roughness

Correct Answer: B) Orientation at a specified angle

Explanation:

Angularity controls orientation relative to a datum at angles other than 90°.

129. Which symbol is NOT a location tolerance?

- A) Position
- B) Symmetry
- C) Concentricity

D) Parallelism

Correct Answer: D) Parallelism

Explanation:

Parallelism is an orientation tolerance, not a location tolerance.

130. Total runout controls:

A) Only circularity

B) Entire surface variation during rotation

C) Hole position only

D) Surface roughness

Correct Answer: B) Entire surface variation during rotation

Explanation:

Total runout controls the cumulative variation of a rotating surface.

131. Which category does the symmetry symbol belong to?

A) Form

B) Orientation

C) Location

D) Profile

Correct Answer: C) Location

Explanation:

Symmetry controls the balanced location of features about a datum plane or axis.